26900153 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCSC301 - OPERATING SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer questions

Answer all questions

Each question carries **2** marks

1. What is a Command Interpreter? [U] / [CO1]
2. Define response time in CPU scheduling. [U] / [CO2]
3. Explain how a binary semaphore provides mutual exclusion between two concurrent processes. [U] / [CO3]
4. Identify the two major methods of recovering from deadlock. [K] / [CO3]
5. In linked allocation, if the first block of a file is known, how can the next block be located? [A] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Outline the advantages and disadvantages of using a Command-Line Interface. [U] / [CO1]

7. Compare batch operating system and time sharing operating system [K] / [CO1]

8. Demonstrate how pipes can be used to transfer data between two related processes. [U] / [CO2]

9. Explain how the PCB is used during a context switch. [A] / [CO2]

10. Explain the role of the producer, consumer, and bounded buffer in the Producer–Consumer Problem. [U] / [CO3]

11. Summarize the role of hardware support in solving synchronization problems. [U] / [CO3]

12. Compare resource preemption and resource ordering as methods of deadlock prevention. [An] / [CO3]

13. The available memory holes are 100 KB, 500 KB, 200 KB, 300 KB and 600 KB. Three processes of sizes 180 KB, 250 KB and 450 KB are to be allocated in the given order. Apply the first-fit, best-fit and worst-fit strategies and determine the memory block allocated to each process. [A] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Explain the long-term, short-term and medium-term schedulers in detail. Discuss their functions, frequency of operation, scheduling decisions . [U] / [CO2]

15. a) Describe the data structures used by Banker's Algorithm and explain their purpose. b) Explain the steps followed by the Banker's Safety Algorithm to determine whether the system is safe. [U] / [CO3]

16. A logical address in a segmentation system is given as 2400. The segment table contains the following entries: [A] / [CO4]

Segment	Base	Limit
0	1000	500
1	4000	800
2	6000	1000
3	8000	600

Use the segment table to determine whether the given logical address is valid and, if valid, calculate its corresponding physical address. Explain the steps involved in the address translation.

[15x2 = 30]